Rider Behavior Analysis

Cyclistic Bike-Share Case Study

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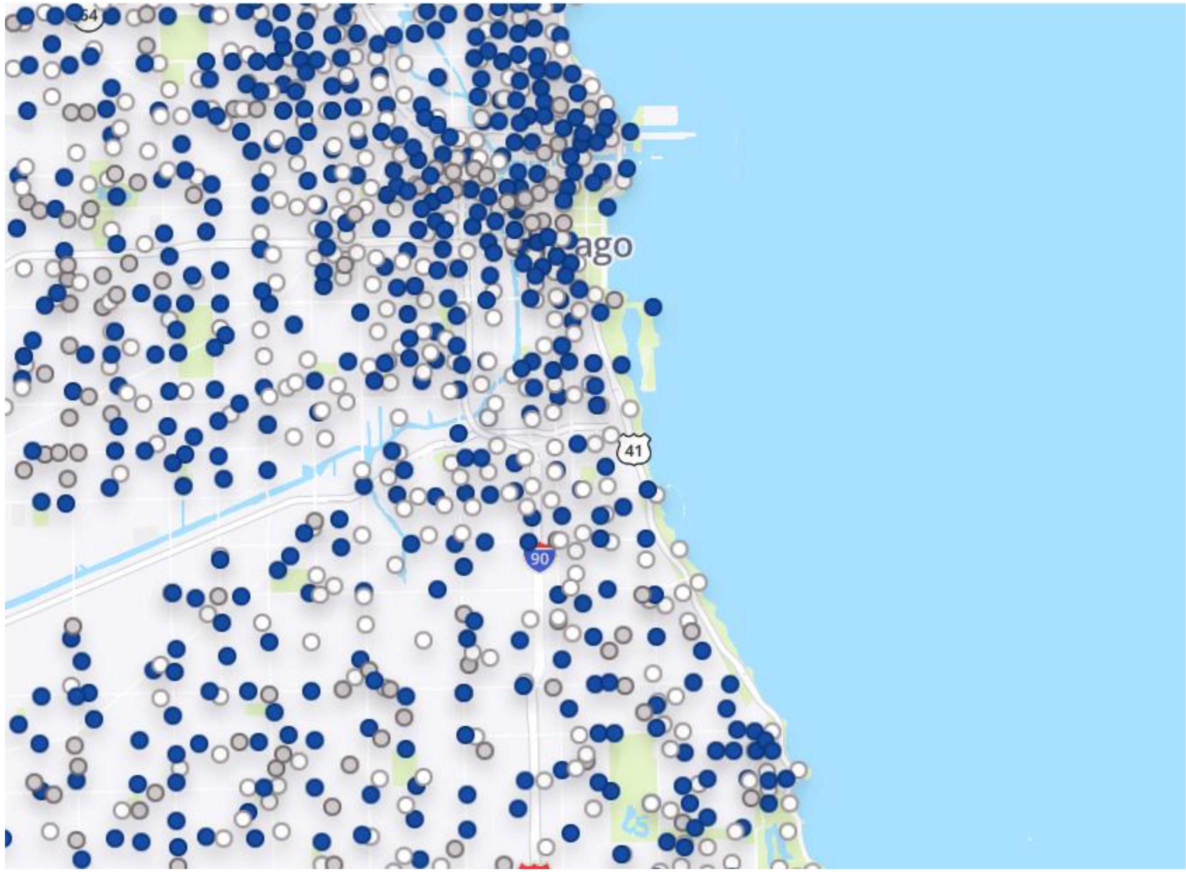
Background

Overview

Cyclistic is a premier bike-share program in Chicago featuring 5,800+ bicycles across 600+ docking stations. It has two types of riders with following pricing plans:

- **Casual Riders:** Single ride or full-day pass users.
- **Annual Members:** Long-term subscription holders.

Cyclistic wants to grow its annual memberships as they are more profitable than casual rides.



Objectives

Goal

Provide insights to support a marketing strategy that converts casual riders into members.

Task

Analyze customer interaction history and purchase behavior data.

Data Source

Provenance

Source: Motivate International Inc. via the divvy-tripdata public bucket.

Scale

Period: May 2025 – May 2026. Consists of 13 months data (13 CSV files with 6,351,159 rows).

Limitations

Privacy: No personal identifiable information. Analysis is restricted to trip behavior; we cannot link rides to individual users.

Methodology

Technical Tools

- **Python:** Primary language for data processing
- **Pandas:** Efficient cleaning and manipulation.
- **Tableau Public:** High-impact visualization.

Core Data Focus

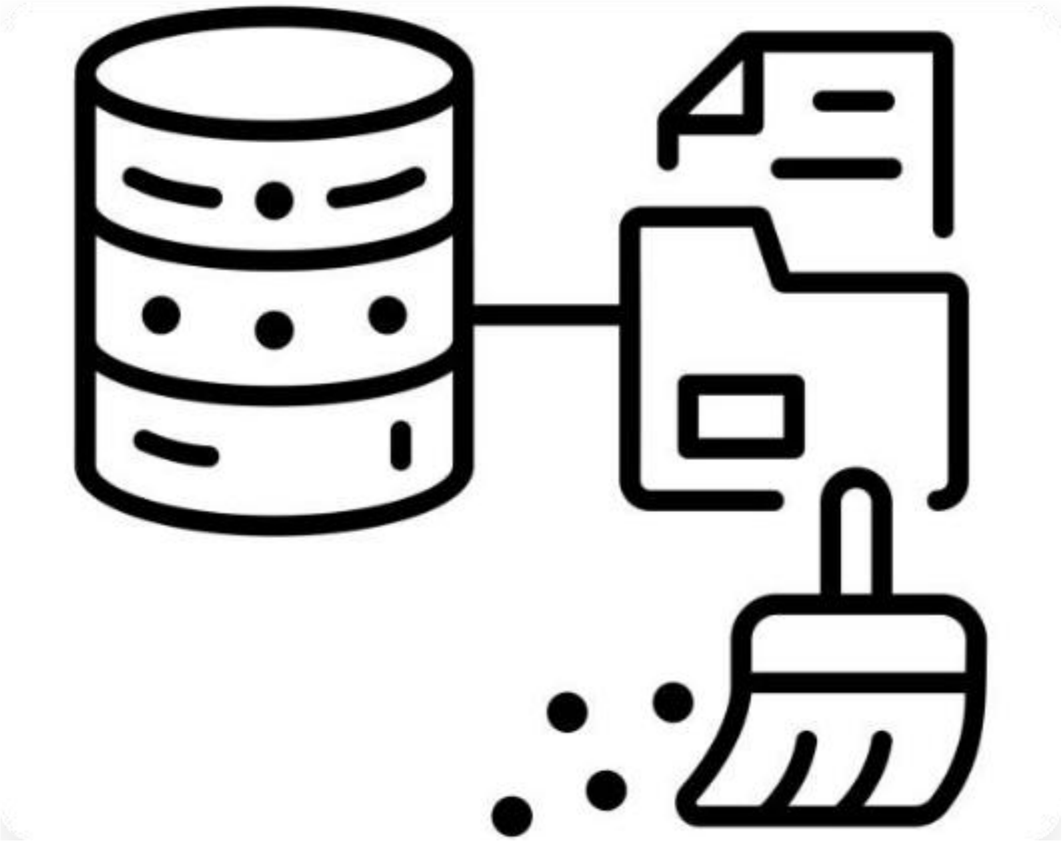
The analysis prioritized three existing primary columns to define rider archetypes and behavioral segments:

started_at ended_at member_casual

Data Cleaning summary

- Removed **6,447 rides** with missing End latitude and End longitude
- Created column **station_name_status** that tracks station name entry
- Engineered column **ride_length**, **hours**, **day_of_week**, and **months**
- Removed **29 rides** with duplicate ride_id
- Removed **124,904** rides with ride_length of negative, zero, less than 1 min and greater than 48 hours

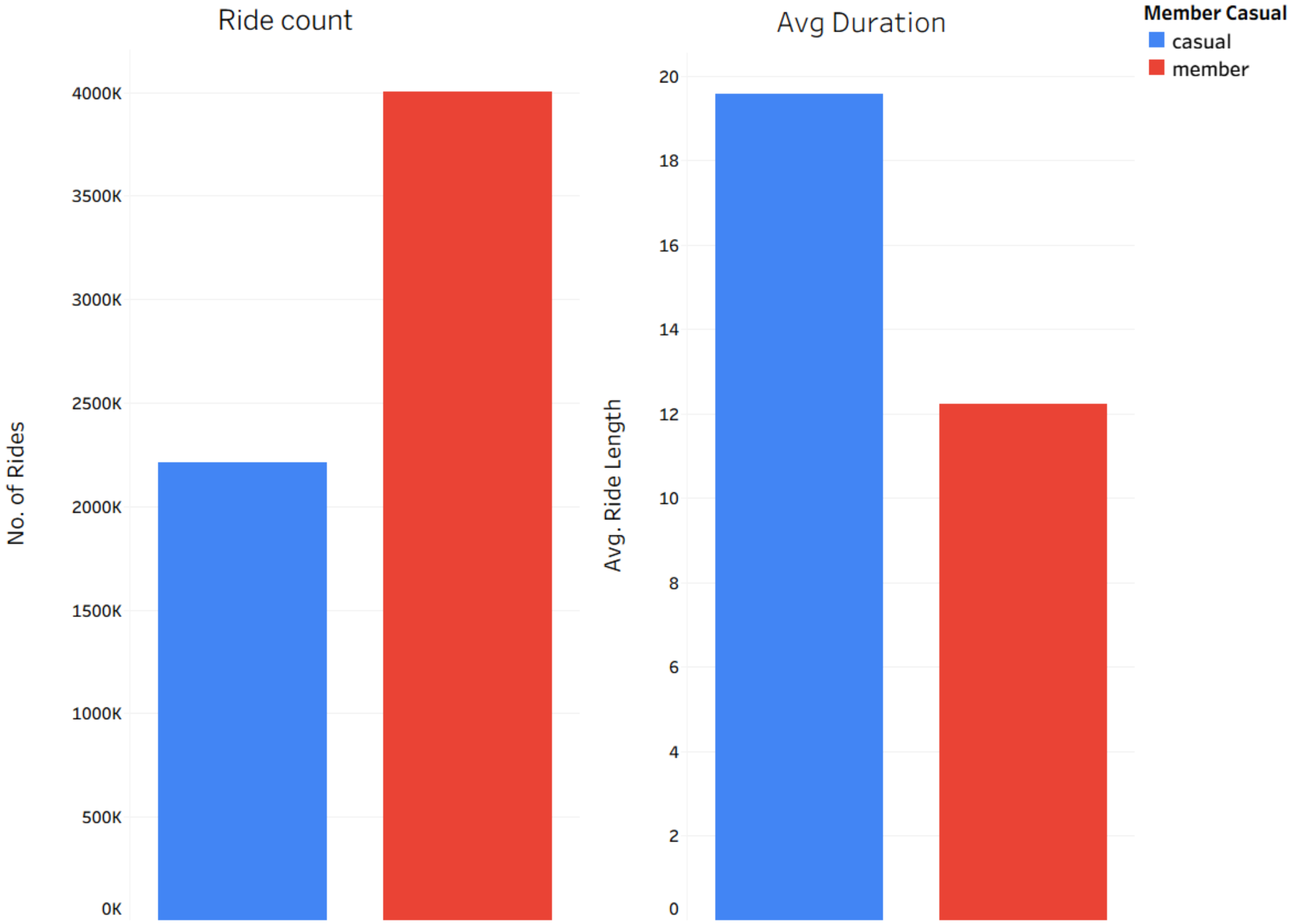
Final Dataset: 6,219,779 Valid Rides



Ride frequency vs Average Duration

Casual riders tend to ride less but for longer duration

- **Members:** Take almost double the total number of rides, indicating regular point-to-point utility or commuting.
- **Casual Riders:** Ride nearly twice as long per trip, indicating leisure, recreation, or weekend tourism.

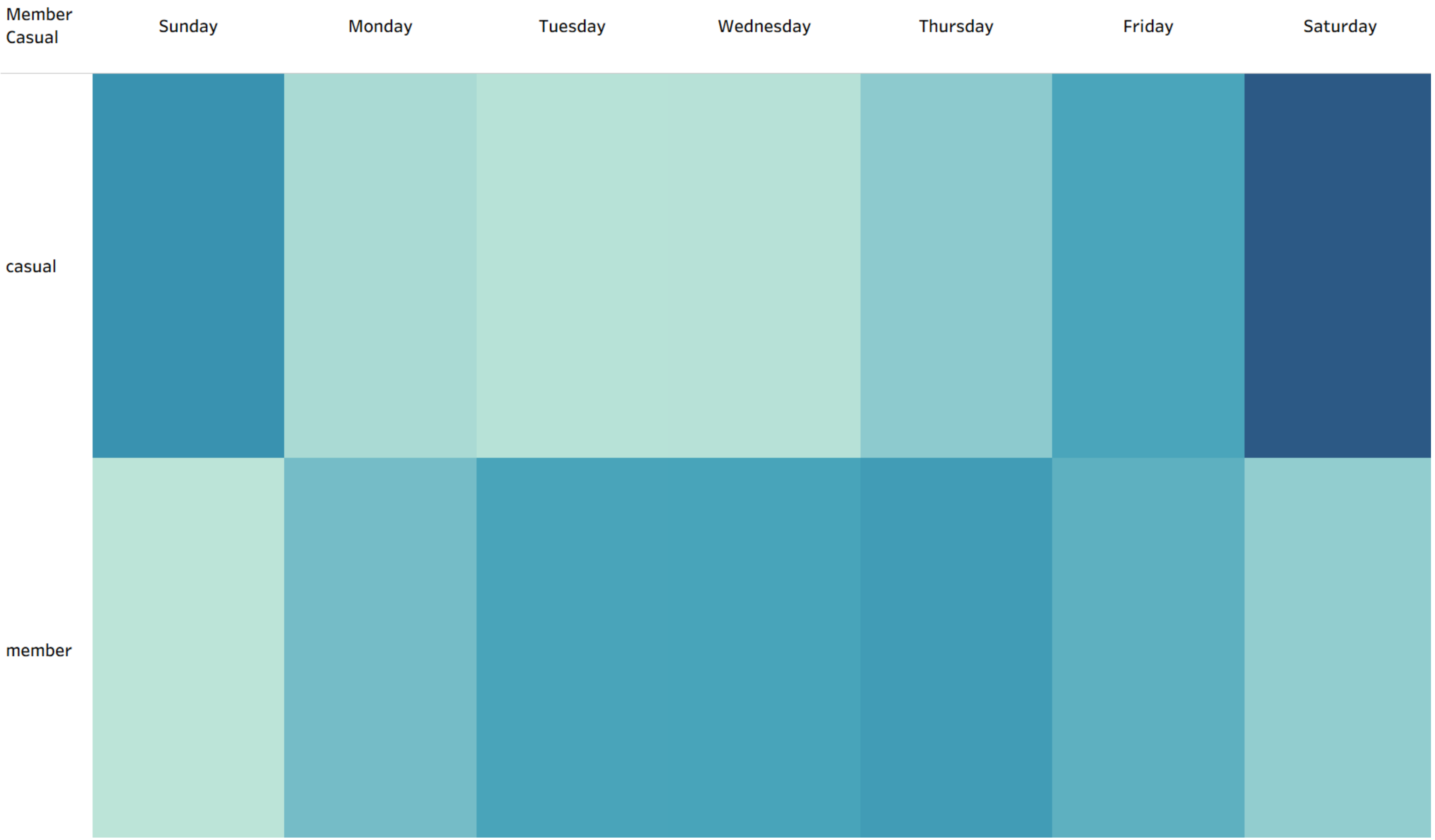


Weekly Ride Distribution

- **Casual Riders** tends to concentrate their rides on weekends

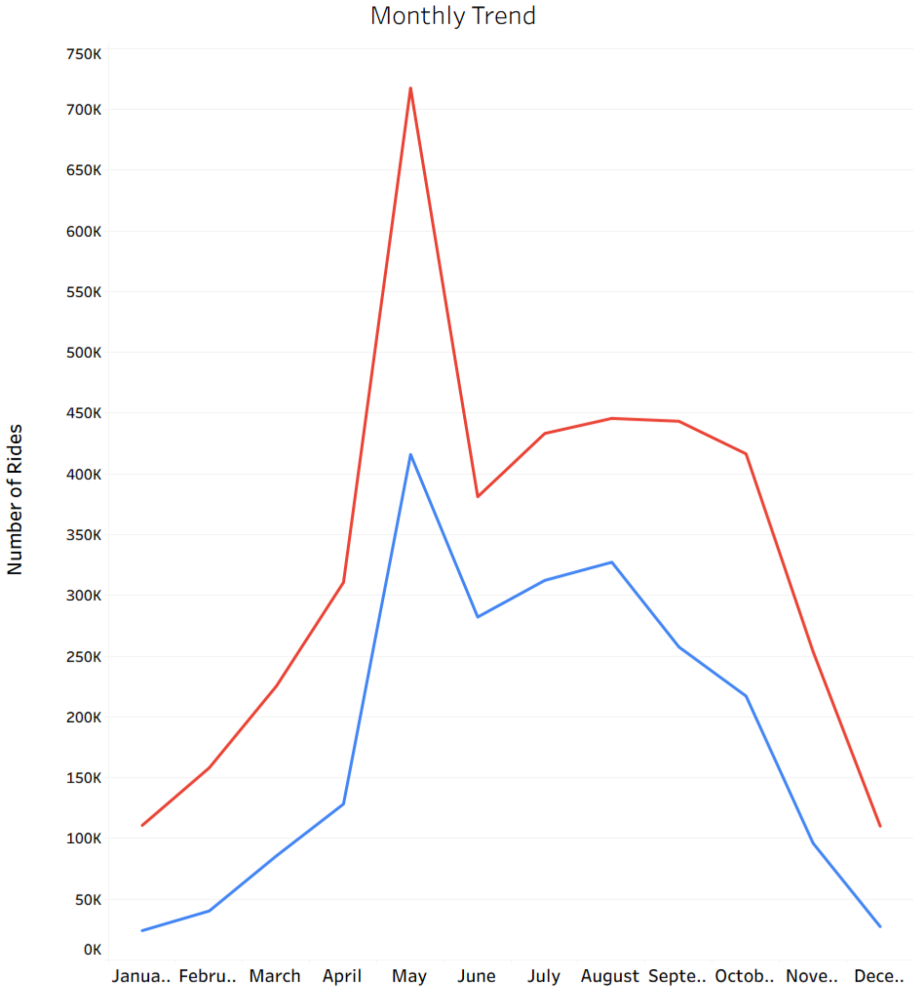
- **Members** tends to spread across the week.

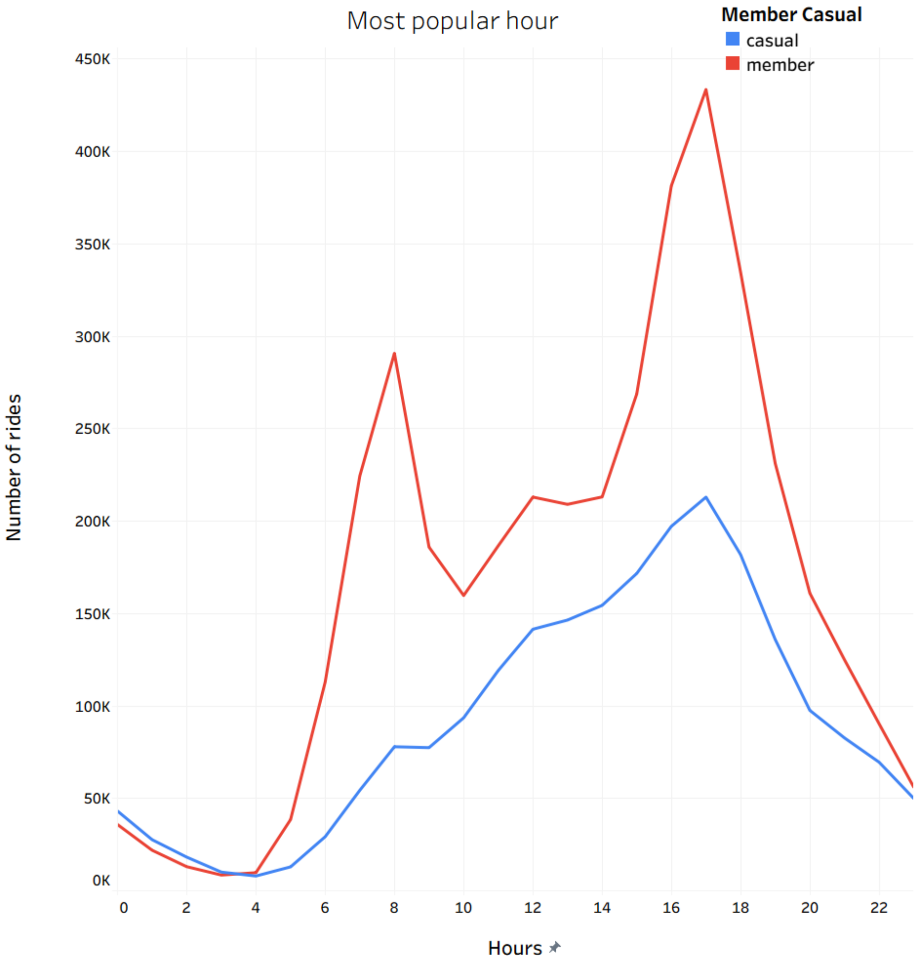
This shows members use bike for daily or routine level travel



Seasonal Surges

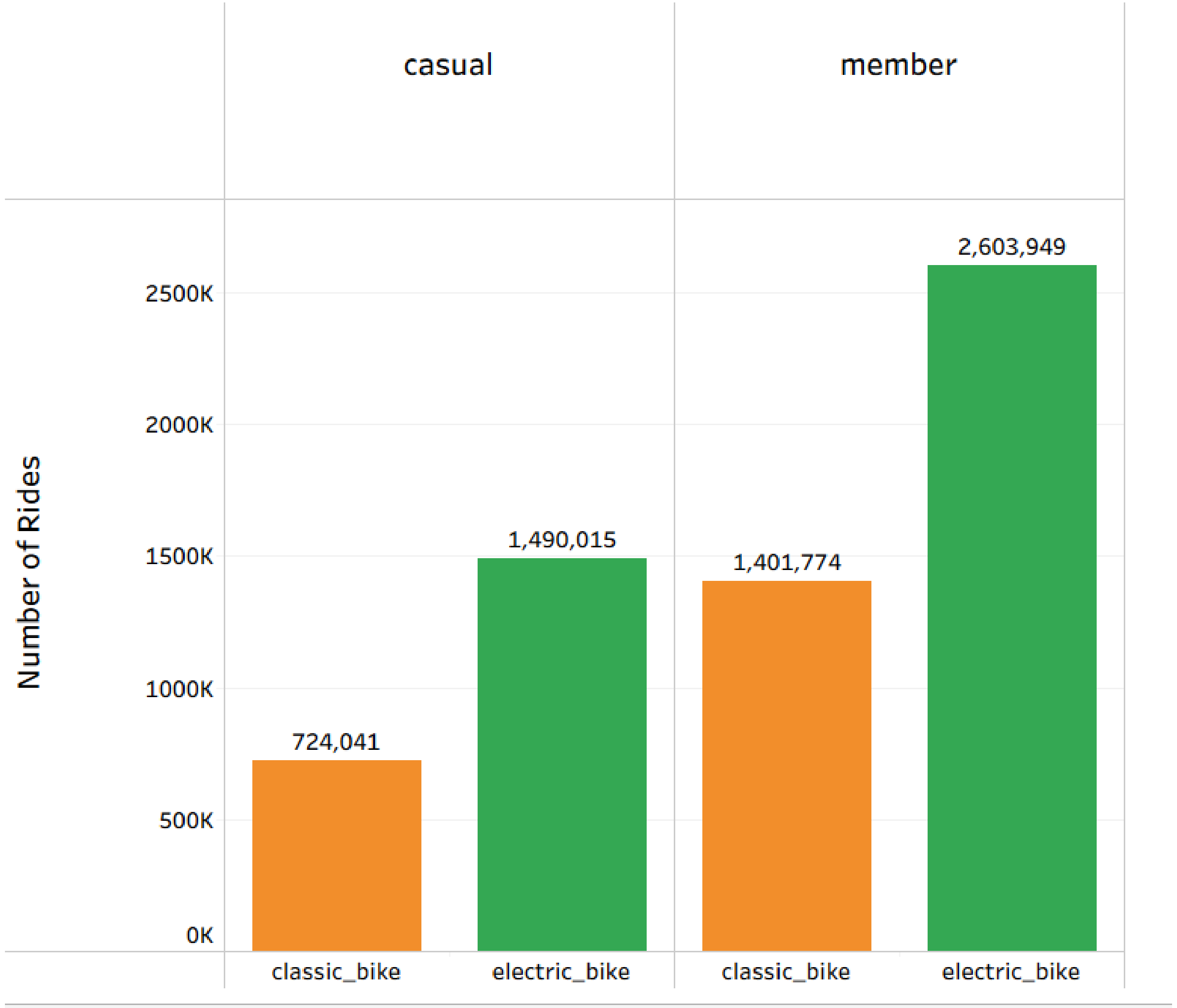
Both rider types most frequently use bikes in May at around 5:00 PM.





Bike preference

Electric bikes are more popular among members than casuals.

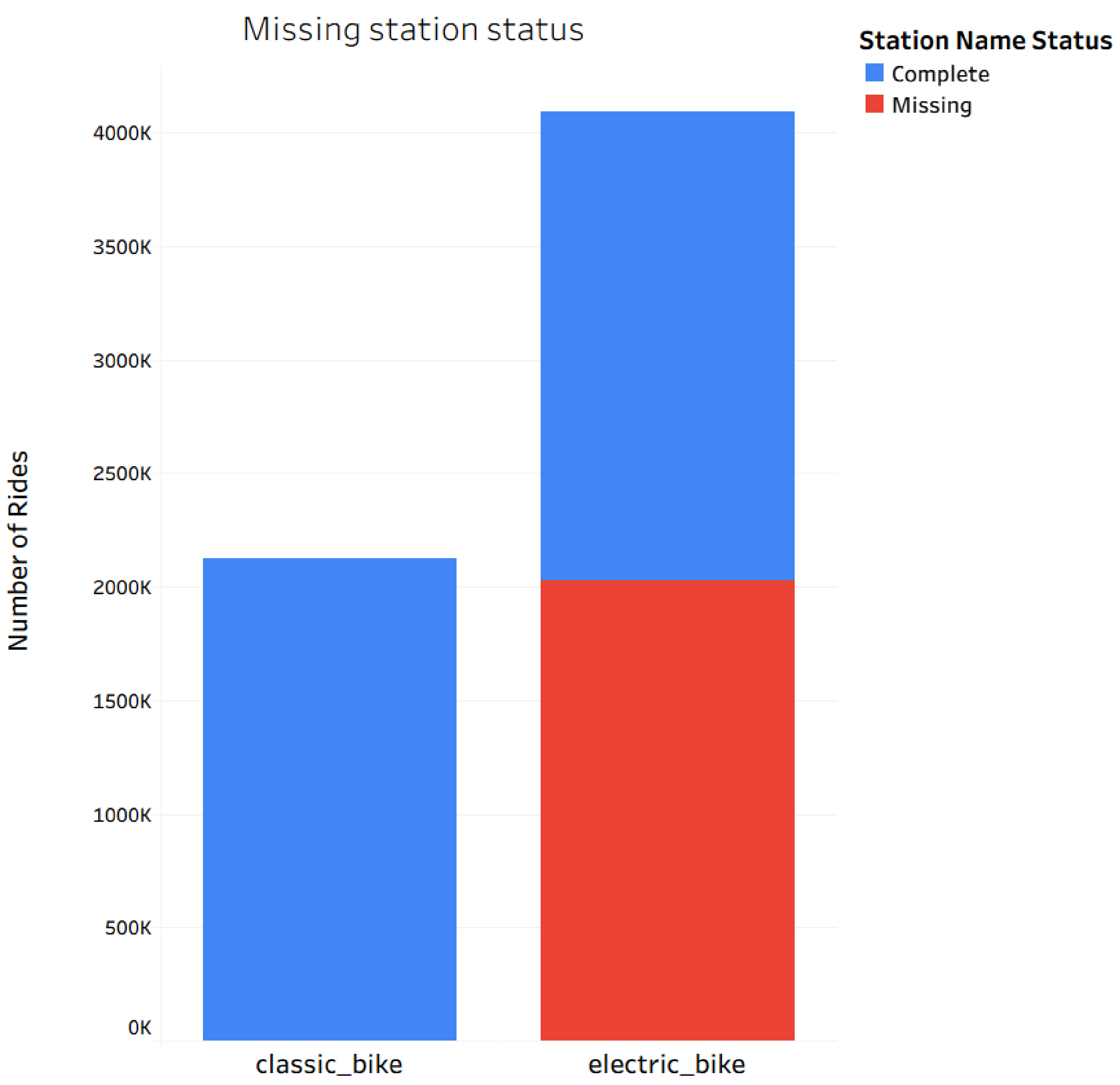


- Electric bikes are the preferred choice for both rider groups.
- Members riding electric bikes make up the largest single segment.
- The ease of electric might be the reason behind people preference

Surprising information

Classic bikes have almost no missing station names

- ➢ The Classic bike shows only blue color, while the Electric bike split nearly 50/50 between complete and missing.
- ➢ This shows many of electric bike can be parked anywhere because of their built-in lock, and this might be another reason behind most people using electric bikes



Strategic Recommendations

Introduce a Weekend Pass Membership

Weekend Pass membership can appeal to casual riders, whose usage peaks massively on Saturdays and Sundays.

Electric Bike Summer Promotion

Launch a summer marketing campaign heavily featuring electric bikes, as casual riders show a strong preference for this rideable type during peak months like May

Implement Ride to Own Credit System

Implement a targeted program where casual riders can apply their accumulated single ride or daily pass fees toward the cost of an annual membership. Show yearly savings for membership upgrades.